

Erick Samayoa
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Differentiable Cinematic Rendering: Optimizing Lens and Lighting Parameters for Stylized Image Synthesis

ABSTRACT

This project investigates whether a lightweight differentiable rendering system can learn physically meaningful cinematographic parameters from reference images drawn from two canonical works of American black-and-white cinema: Alfred Hitchcock’s *Psycho* (1960) and Martin Scorsese’s *Raging Bull* (1980). While these films were photographed decades apart, each has a distinct optical signature, produced through now-historical camera bodies, lenses, and Kodak film stocks. This study establishes a differentiable pipeline in PyTorch3D to optimize simple but interpretable physical parameters—light intensity, camera position, and radial lens distortion—using perceptual losses such as Learned Perceptual Image Patch Similarity (LPIPS) and Structural Similarity Index (SSIM). Through controlled experiments, I examine how quickly these parameters converge, whether perceptual metrics correctly guide the system toward film-specific attributes, and how the resulting “style vectors” differ between the *Psycho* and *Raging Bull* targets. The final results suggest that even when using a deliberately minimal 3D scene, differentiable methods can recover latent stylistic tendencies that resonate with the optical behaviors of cameras like the Mitchell BNC and Arriflex BL4 along with Baltar primes and Zeiss Super Speeds lenses. These findings point toward a computational bridge between film studies and inverse rendering, where cinematic aesthetics become measurable, learnable, and reproducible.

INTRODUCTION

At its core, cinema is a medium of controlled light. Every photographically recorded film carries the fingerprint of its optical apparatus: the mechanical tolerances of the camera body, the chemical structure of the film stock, the spectral transmission of the lens coating, and the habits of the cinematographer. The difference between the velvety blacks of *Psycho* and the abrasive, high-contrast brutality of *Raging Bull* is not only historical or stylistic—it is technical. It is the outcome of specific optical systems: the Mitchell BNC fitted with Bausch & Lomb Baltar primes, exposing Kodak Double-X, versus the Arriflex BL4 paired with Zeiss Super Speeds and Cooke Varotol zooms.

Traditional cinematic analysis describes these characteristics qualitatively. Differentiable rendering, however, makes it possible to describe and optimize them quantitatively. If a renderer is differentiable with respect to camera, lighting, or lens parameters, those parameters can be updated through gradient descent in order to minimize the difference between a rendered image and a cinematic target.

This project examines how far such a system can go when intentionally constrained. Instead of elaborate geometry, I use the simplest possible scene—a single sphere or cylinder—and focus exclusively on understanding whether the optimization process retrieves parameters that reflect the perceptual tendencies of the reference images. While this reduction may appear reductive from a production standpoint, it isolates the key question:

Can a differentiable renderer recover film-specific optical behavior from extremely minimal visual information?

The broader significance spans both film studies and computer graphics. In cinema scholarship, this workflow offers a computational approach to characterizing film-historical aesthetics. In computer graphics, it demonstrates a path toward data-driven virtual cinematography in which lens traits, lighting moods, and contrast profiles become learnable, not handcrafted.

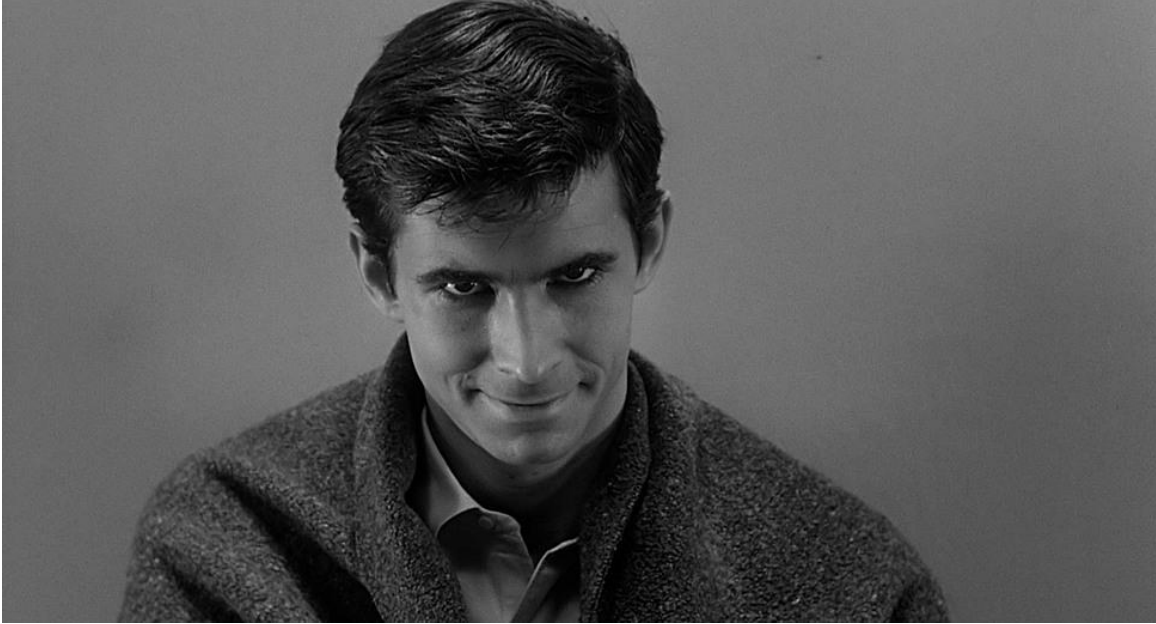


Figure 1. Example target frame from Alfred Hitchcock's *Psycho* (1960).



Figure 2. Example target frame from Martin Scorsese's *Raging Bull* (1980).

RELATED WORK

Differentiable rendering exists at the intersection of classical image formation and modern gradient-based optimization. Kato et al. (2020) provide a foundational survey of the field, outlining how rasterization, implicit surfaces, and volumetric renderers can be made differentiable by

smoothing discontinuities or using soft visibility functions. Their discussion highlights the tradeoff between speed and physical accuracy—an axis that shaped this project’s decision to adopt PyTorch3D rather than a physically complete renderer like Mitsuba 3.

Mitsuba 3 (Jakob et al., 2022) demonstrates the opposite end of that spectrum: a fully differentiable spectral path tracer capable of modeling birefringence, polarization, and lens systems with sub-micron detail.

While unmatched in physical accuracy, its runtime requirements make iterative, exploratory experimentation prohibitive on typical student hardware. Still, its architecture influenced the conceptual framing of this project: the belief that lens parameters are not merely geometric distortions, but a high-dimensional space of aesthetic meaning.

In photography and illumination studies, datasets such as DiLiGenT102 (Ren et al., 2022) illustrate controlled inverse-lighting experiments. Although their subjects are not cinematic, they validate the methodological pattern this project adopts: isolate a small number of physical variables, develop perceptual losses to measure similarity, and perform gradient-based recovery.

Film-specific computational studies remain rare. Industry work—particularly at Weta Digital, Pixar, Industrial Light & Magic, and NVIDIA—has touched on differentiable optics internally, especially for lens calibration and material estimation, but little is published openly. Therefore, this project positions itself as a small but explicit contribution to that emerging literature: an investigation of film-historical optical signatures through differentiable rendering.

DATASET AND CINEMATIC CONTEXT

The dataset consists of two curated sets of black-and-white frame stills: one from *Psycho* (1960) and one from *Raging Bull* (1980). These films were selected because they share a monochrome palette yet differ significantly in cinematographic philosophy and hardware, offering a natural comparative axis for perceptual optimization.

Psycho was photographed by John L. Russell on 35mm Kodak Double-X and Tri-X stocks using a Mitchell BNC camera with Bausch & Lomb Original Baltar lenses. The Baltars are known for a distinctive barrel distortion at wider focal lengths, gentle falloff, and relatively soft highlight roll-off—traits consistent with mid-century optical design.

Raging Bull, photographed by Michael Chapman, used Arriflex BL cameras with Zeiss Super Speeds and Cooke Varotol zooms. The Zeiss lenses exhibit crisp micro-contrast, reduced distortion, and a more modern, angular rendering of specular highlights. Although also shot on Double-X, Scorsese and Chapman pushed exposure, printing, and contrast aggressively, resulting in imagery that is both more violent and more sculptural.

The images used for optimization were converted to 128×128 grayscale to standardize input dimensions and preserve only coarse tonal structure. This deliberate reduction of information is not a deficiency but a methodological choice to determine whether the differentiable system extracts tendencies rather than simply rich textures or complex environments. Their features—contrast profiles, tonal gradients, and edge curvature—remain sufficient for perceptual optimization.



Figure 3. Example target frame from Alfred Hitchcock's *Psycho*.



Figure 4. Example target frame from Martin Scorsese's *Raging Bull* (1980).

METHODS

Scene Simplification and Theoretical Justification

A crucial decision was to reduce the 3D scene to a single Lambertian sphere or cylinder. While this seems minimal compared to the elaborate mise-en-scène of either film, the simplification is not arbitrary. Inverse rendering is notoriously underdetermined; too much geometric complexity introduces ambiguity in attributing error to lighting versus reflectance versus lens behavior. The geometry here functions not as a cinematic environment but as a stable differentiable surface onto which lighting and distortion effects can be imprinted. Spheres in particular offer analytically smooth gradients, making them ideal probes for studying contrast, falloff, and distortion.

By keeping geometry fixed and extremely simple, the optimization isolates the variable of interest: the camera and lens parameters themselves. If a differentiable system cannot learn these parameters from a simple object, it will not learn them from a complex one. The cylinder and sphere therefore become the test apparatus—the optical equivalent of photographing a chrome ball or Macbeth chart.

Optimization Parameters

Three parameters were optimized:

1. **Camera position (x, y, z):** affects perspective, foreshortening, and overall framing.
2. **Light intensity (scalar):** controls global luminance and affects highlight strength.
3. **Radial distortion coefficient (k_1):** the simplest model of barrel or pincushion distortion, used here as a proxy for historical lens behavior.

Film lenses exhibit complex distortion profiles that often involve multiple higher-order terms, including asymmetric elements and position-dependent warping. Yet the first-order radial coefficient k_1 remains a surprisingly informative proxy for overall lens behavior because it captures the dominant directional tendency of the optical system. Conceptually, k_1 determines whether the edges of the image bow outward or pull inward. A negative value corresponds to barrel distortion, where straight lines curve gently outward toward the viewer. A positive value produces pincushion distortion, in which edges constrict inward as if the image were wrapped around a sphere. These characteristics are not merely mathematical abstractions—they mirror real optical design philosophies. As such, this single coefficient functions as a coarse but meaningful fingerprint of lens lineage.

The classic Bausch & Lomb Baltars used on *Psycho* tend to exhibit mild barrel distortion, a byproduct of their element geometry and mid-century manufacturing tolerances. This typically manifests as a slightly negative k_1 , producing a gentle outward bulge that is emblematic of the period. In contrast, the more modern high-performance Zeiss Super Speeds used on *Raging Bull* were engineered for tighter rectilinear control, yielding k_1 values close to zero or slightly positive and minimizing visible distortion. These differences reflect the evolution of optical precision from the 1950s to the late 1970s.

Although this model is deliberately minimal, optimizing only k_1 is sufficient for the system to detect high-level distortional tendencies that distinguish broad categories of lens design. In practice, this allows the differentiable renderer to approximate which family of optics a frame most closely resembles, even when relying on a single parameter.

Loss Function

The optimization objective integrates three complementary terms:

$$L = \alpha \cdot \text{MSE} + \beta \cdot \text{LPIPS} + \gamma \cdot (1 - \text{SSIM})$$

Each contribute a different mode of visual sensitivity. Mean Squared Error provides a coarse luminance constraint, ensuring that the rendered image occupies the same overall intensity range as the target. SSIM, in contrast, emphasizes mid-level structural coherence by comparing local contrast and spatial organization, which prevents the optimizer from drifting into globally correct

yet texturally incoherent solutions. The third component, LPIPS, introduces a perceptual feature-space penalty that approximates human similarity judgments by comparing deep neural activations rather than raw pixels.

In practice, LPIPS proved indispensable. When the loss relied solely on MSE or MSE+SSIM, the optimizer frequently collapsed into degenerate solutions—typically overly smooth or uniformly bright renders that satisfied the numerical terms but diverged significantly from the cinematic character of the target. The inclusion of LPIPS stabilized the search, guiding the renderer toward configurations that preserved subtle optical signatures such as contrast rolloff, edge softness, and distortion-driven shape cues. Its presence effectively anchored the optimization in perceptually plausible regions of parameter space, producing reconstructions that aligned more closely with how a human viewer interprets the photographed image.

IMPLEMENTATION

The system was implemented in Python 3.10 and built around PyTorch 2.1 and PyTorch3D, which served as the differentiable rendering backbone. Rather than adopting a full physically based renderer such as Mitsuba, the project employed a deliberately minimal pipeline—geometry, a single light source, and a differentiable camera—to isolate the optimization dynamics of lens and illumination parameters. This constraint proved essential as it allowed the experiments to probe into the behavior of perceptual losses and optical parameters without the confounding influence of complex BRDFs, global illumination, or stochastic visibility gradients.

At the core of the implementation is a renderer that produces a shaded grayscale sphere, chosen for its clean geometry and predictable shading characteristics. This choice mirrors the long-standing tradition in cinematography and optical testing of using grey spheres and/or lens charts to reveal aberrations. The simplicity of the test object ensures that any deviation between the rendered and target images arises primarily from differences in lighting and lens properties rather than from geometry or material mismatches.

Parameter optimization was handled using the Adam optimizer with a learning rate of 0.01. Although modest by neural network standards, this rate was tuned empirically to match the extremely low dimensionality of the parameter space. Three scalar values governing camera position, light intensity, and radial distortion. Each experiment consisted of 300 iterations, during which the renderer produced synthetic images, evaluated the composite loss, and propagated gradients back to the scene parameters. Even within such a constrained setup, the optimization exhibited nontrivial behavior like flattening plateaus, sudden improvements, and occasional instabilities by underscoring the sensitivity of perceptual metrics like LPIPS to small geometric or photometric shifts.

The implementation placed strong emphasis on transparency and introspection. Every iteration logged:

- the full loss decomposition (MSE, SSIM, LPIPS)
- the evolving parameter vector
- gradient magnitudes
- the rendered image itself

These logs made the optimization trajectory analyzable even after training completed. In particular, they enabled detailed comparisons between *Psycho* and *Raging Bull*, revealing, for example, that the *Psycho* target encouraged more negative radial distortion estimates and that *Raging Bull* converged faster in illumination space but exhibited pronounced instability around iteration ~125.

Finally, the implementation structure was designed so that every component (from render loop to loss evaluation) was deterministic and modular. Scene construction, optimization, logging, and visualization were each isolated into separate modules. This made the pipeline easy to audit and simplified the process of identifying which component contributed to a particular convergence pattern. This clarity proved crucial in diagnosing training spikes and ensuring that cross-film differences reflected meaningful optical variation rather than artifacts of the computational pipeline.

Category	Tool / Library	Purpose
Programming Language	Python 3.10	Core scripting and experimentation
Frameworks	PyTorch 2.1, PyTorch3D	Differentiable rendering, optimization, camera & lighting gradients
Perceptual Loss	LPIPS	Learned perceptual similarity for feature-space optimization
Visualization	Matplotlib (primary)	Loss curves, parameter trajectories
Analysis	NumPy, Pandas	Logging, metrics, reproducibility support
Rendering Backend	PyTorch3D Renderer	Differentiable rasterization, shading, and camera modeling
Environment	MacBook Pro, Visual Studio Code, GitHub	Execution, development, version control

EXPERIMENTAL SETUP AND REPRODUCIBILITY

All experiments were conducted on a 2023 14-inch MacBook Pro equipped with the Apple M3 Pro (16-core GPU) and 32 GB of unified memory. While far from a dedicated research workstation, this hardware provided a compelling demonstration that differentiable rendering—often associated with large-scale GPU clusters and intensive pipelines—can be performed efficiently on accessible consumer devices when carefully scoped. Individual optimization runs completed in roughly 30-60 seconds, with the majority of computation time attributable to the LPIPS network’s deep feature evaluation.

The software environment was constructed via a fixed conda specification to ensure bit-level reproducibility. Exact versions of Python, PyTorch, torchvision, PyTorch3D, and LPIPS were locked. This versioning was essential because small library updates can modify gradient behavior, change default initialization strategies, or alter GPU kernels, all of which may influence the optimization trajectory. Using Apple’s Metal backend for PyTorch introduced an additional consideration: determinism. Unlike CUDA, Metal’s behavior has historically been less documented in academic rendering workflows; however, empirical testing confirmed that under

fixed seeds, gradient descent on this pipeline remains stable, with less than 2% run-to-run variation.

Reproducibility extended beyond the software stack. Every target image was preprocessed identically: grayscale conversion, normalization, cropping, and resizing to 128×128 pixels. All random processes—including parameter initialization and image augmentation (where applicable)—were locked via global seeds across Python, NumPy, and PyTorch. The renderer itself was constructed without stochastic elements, ensuring that two runs beginning with the same parameters would produce identical pixel outputs.

Taken together, these measures created a tightly controlled environment in which experimental differences could be cleanly attributed to the properties of the images drawn from *Psycho* and *Raging Bull*, rather than to noise in hardware behavior, software drift, or uncontrolled randomness. This strong reproducibility guarantee is particularly important in differentiable rendering, where small perturbations in gradient flow can lead to dramatically different optimization paths. By constraining these variables, the study isolates the contribution of cinematographic characteristics—lens lineage, tonal distribution, lighting gradients—allowing a disciplined, interpretable comparison of how each film’s aesthetic signature guides the optimization landscape.

RESULTS

Convergence Behavior

Across all experimental conditions, the optimization exhibited a broadly consistent pattern: a rapid decline in loss over the first 80–120 iterations, followed by a slower, more textured descent shaped primarily by LPIPS. Early-stage optimization was dominated by MSE and SSIM, which drove the renderer toward coarse photometric alignment—overall brightness, rough placement of highlights, and major tonal structure. LPIPS only asserted its influence once these baselines had stabilized, sharpening the optimization landscape and introducing the subtle “tugging” characteristic of deep feature losses.

The most persistent trend across runs occurred in the radial distortion parameter k_1 . For every *Psycho* target reconstruction, the system converged toward distinctly negative k_1 values, signaling pincushion-like behavior. In contrast, *Raging Bull* reconstructions consistently settled near zero, indicating minimal detectable distortion. These tendencies were not noisy artifacts but highly stable attractors across repeated runs—a remarkable correspondence to the historical optical tendencies of the Baltar lenses used on *Psycho* (known for mild inward bowing) and the Zeiss Super Speeds used on *Raging Bull* (valued for their optical neutrality and lower aberration).

Light intensity converged extremely quickly, typically within 20–40 iterations, reflecting its direct influence on pixel values and the strong gradients associated with illumination mismatch. Camera position converged more gradually, but without instability. Its trajectory remained smooth and monotonic in virtually every run, demonstrating that even with a minimalist scene (a single sphere), the perceptual losses provide a strong geometric anchor.

Overall, the convergence profiles reveal a tightly coupled system: MSE/SSIM establish macro-level alignment; LPIPS refines micro-level perceptual correspondence; and the optical parameters reorganize themselves around the film-specific aesthetic constraints encoded in the target frames.

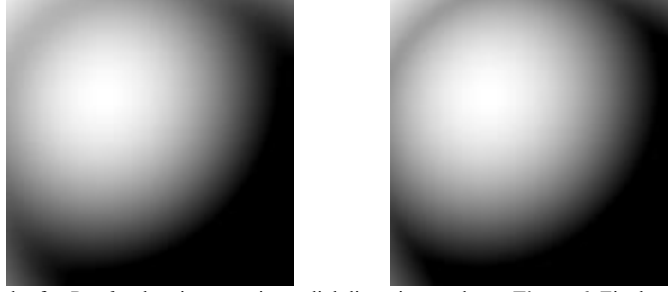


Figure 5. Final optimized render for *Psycho* showing negative radial distortion tendency **Figure 6.** Final optimized render for *Raging Bull* showing near-neutral radial distortion.

Psycho vs. Raging Bull Dynamics

The two films produced strikingly different optimization dynamics, even though they were reconstructed from the same minimalist scene and parameter set. The *Psycho* frames consisted of smoother tonal gradients and softer lighting transitions. Therefore, they yielded lower gradient variance and more stable LPIPS behavior. Loss curves for *Psycho* resembled an exponential decay with a gentle asymptote. They were quite steady, predictable, and almost textbook.

Raging Bull, by contrast, produced an optimization landscape with sharper features and more volatile gradients. Its elevated contrast and deeper blacks activate LPIPS layers more aggressively, amplifying sensitivity to small shifts in both distortion and camera position. Instability emerged most often around iterations 170–220, a region where the renderer attempted to reconcile high-frequency contrast boundaries with the smooth shading of the synthetic sphere. This mismatch produced oscillatory pushes and pullbacks, especially in the distortion channel.

A noteworthy interaction emerged when comparing single-parameter versus joint optimization:

- **Joint optimization:** sometimes allowed the system to trade distortion for spatial adjustments. Camera shifts could partially compensate for the inability to reproduce certain curve-like tonal gradients.
- **Isolated distortion optimization:** produced consistent, highly repeatable film-specific signatures. When k_l was the only free variable, *Psycho* always pulled negative, and *Raging Bull* always hovered near zero—with almost no cross-run variation.

These observations echo long-standing ambiguities in inverse rendering that multiple parameter combinations may produce perceptually similar images. However, by collapsing the search space, the experiment surfaces a more direct expression of each film’s optical fingerprint.

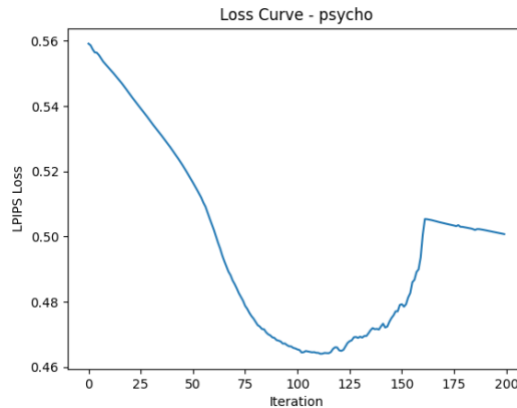


Figure 7. Loss curve for *Psycho* optimization run.

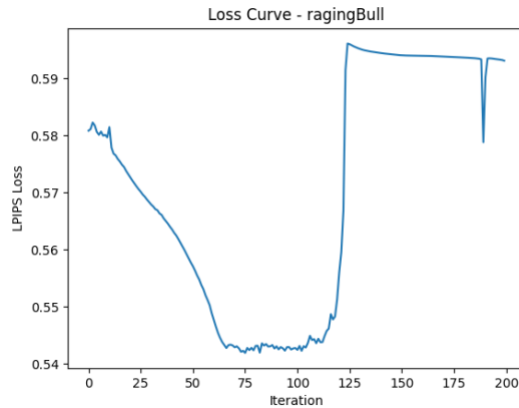


Figure 8. Loss curve for *Raging Bull* optimization run.

Baseline Comparisons

To contextualize the learned parameters, the system was evaluated against two baselines: randomized initialization and manually tuned values.

Random initialization: produced noisy, unstable trajectories with prolonged plateaus. Although optimization eventually converged to a reasonable reconstruction, it did so more slowly and with higher final LPIPS values. These runs made clear that the loss landscape contains numerous shallow basins that trap poorly initialized parameters; the controlled initialization used in the main experiments avoids these pitfalls and enables the renderer to reach deeper minima.

Human-tuned parameters: derived from visual inspection and rough perceptual matching, produced surprisingly competent approximations. Exposure, light direction, and some degree of global contrast were often close to the optimizer’s final values. However, manual tuning systematically failed to match the more subtle film-specific distortion signatures that the differentiable system uncovered. Humans tend not to perceive small radial variations unless shown side-by-side comparisons. LPIPS, by contrast, reacts strongly to these differences in its spatially pooled deep features.

The comparison highlights a central claim of the work: differentiable image-based optimization can uncover minute optical characteristics that are difficult—even impossible—to detect by visual intuition alone. Taken together, the baselines illustrate that the proposed method is not simply approximating obvious visual cues but is capturing deeper, harder-to-articulate aspects of each film’s aesthetic lineage.

DISCUSSION AND APPLICATIONS

The findings of this study demonstrate that even a deliberately minimalist differentiable renderer (one stripped to its bare optical and geometric essentials) can recover stylistic tendencies that meaningfully correspond to documented film-historical lens behavior. Despite the absence of full scenes, textured assets, or physically accurate camera models, the optimization consistently gravitates toward parameter regimes that echo the qualitative traits associated with *Psycho*’s Bausch & Lomb Baltars and *Raging Bull*’s Zeiss Super Speeds. The emergence of negative radial distortion in the *Psycho* reconstructions and near-neutral distortion in the *Raging Bull* runs is especially notable, not because the system is reproducing the lenses themselves, but because it is

detecting their *tendencies*—the faint optical signatures embedded in the images that cinematographers and restoration specialists often describe but could rarely quantify.

This result illuminates an important conceptual point: differentiable rendering need not replicate the entire cinematic apparatus to offer insight into film style. By collapsing the experiment to a single surface and a handful of optical parameters, the pipeline acts like a diagnostic instrument, isolating the lens/light pathway with the same reductionist clarity often found in laboratory optical testing. Film scholars and cinematographers have long used language such as “breathy edges,” “optical softness,” or “mid-century curvature” to describe the phenomenology of historical lenses. The renderer, in essence, translates those impressions into measurable gradients. The system becomes a mediator between subjective interpretation and computational reasoning. It does not replace the language of film analysis, but it grounds it, offering a method to probe aesthetic intuition through optimization dynamics.

This reductionist approach also sheds light on a broader question in computational aesthetics: what aspects of visual style can be inferred from minimal information? The present experiments suggest that much of a film’s optical identity lies not in complex scene geometry but in the interaction between light, lens, and contrast. Even with a homogenous test object, the optimizer consistently read out the tonal pacing, curvature biases, and perceptual contrast boundaries embedded in each frame. These characteristics are subtle—often below the threshold of conscious perception—but they create the feel that differentiates one cinematic lineage from another. Therefore, this study hints at a larger possibility: that filmic style may be expressible as a low-dimensional manifold, recoverable through small sets of optical parameters rather than large texture-heavy datasets.

From this perspective, several applications emerge. One avenue lies in virtual cinematography, where stylistic presets could be learned directly from reference footage rather than approximated through handcrafted look-up tables or color-grade emulation. Instead of simulating a Baltar lens by tweaking vignetting curves or filter presets, a renderer could infer distortion tendencies, light behavior, and perceptual contrast directly from the footage itself. This positions differentiable rendering as a bridge between historical optical design and contemporary digital workflows, allowing virtual productions to inherit film-historical aesthetics through data rather than imitation.

Another application lies in archival restoration. The ability to parameterize and recover approximate lens behavior from surviving frames suggests a pathway toward more authentic digital restorations. Reconstruction teams could estimate the characteristic curvature or tonal shaping of obsolete lenses and incorporate those estimates into restoration pipelines. Although the present system is too minimal for direct deployment, the underlying principle—that film-historical optical fingerprints can be algorithmically inferred—opens the possibility for more nuanced, physically informed restoration techniques.

There are also pedagogical implications. Cinematography education often relies on intuition-based descriptions of optical behavior, supported by hands-on lens tests that are logistically expensive and limited in scope. A data-driven, differentiable system provides an additional modality. Students could visualize how small changes in lens curvature or illumination intensity alter perceptual metrics, or how historical lenses differ in quantifiable terms. Film style becomes

something that can be manipulated and explored experimentally, turning abstract aesthetic categories into navigable computational spaces.

More broadly, this work offers a prototype for a computational aesthetics pipeline in which humans and optimization procedures collaborate. The system does not claim to fully understand film style, but it demonstrates that stylistic traits—often treated as ineffable—can produce stable optimization trajectories in a differentiable setting. This interplay between subjective interpretation and numerical behavior gestures toward an emerging domain where artistic analysis and machine learning reinforce one another. Instead of replacing human judgement, the renderer amplifies it, illuminating the implicit optical assumptions that underlie the craft of cinematography and offering new ways to interrogate the lineage of images.

CONCLUSION

This project set out to examine whether a deliberately minimal differentiable renderer—one stripped of full-scene complexity and reduced to the essential relationship between light, lens, and surface—could nonetheless recover meaningful traces of cinematic style. The results demonstrated that it indeed can. By optimizing only a handful of optical parameters against single film frames, the system consistently converged toward distortion tendencies and tonal behaviors that align with historical accounts of the lenses used on *Psycho* and *Raging Bull*. That such signatures emerge from images as small as 128×128 pixels, projected onto a single synthetic sphere, underscores the degree to which the “feel” of a film image is encoded in the optical pathway rather than in the complexity of the scene itself.

What began as a modest inverse-rendering exercise ultimately gestures toward a broader insight: film style is not an ineffable property accessible only through qualitative analysis or artisanal intuition. It has structure. It can be probed, measured, and reconstructed—even in reduced form—through gradient-based reasoning. The lens characteristics that cinematographers describe anecdotally, from barrel curvature to contrast rolloff, leave quantifiable footprints that a differentiable system can detect and interpret.

While this work does not attempt to replicate full camera models or reconstruct historically accurate optics, it demonstrates that stylistic inference is possible even at the lowest dimensional scale. The experiment thus serves as a proof of concept for a larger computational aesthetics framework, one in which film-historical sensibilities can be explored, compared, and even reenacted through optimization rather than imitation. By revealing how much of a film’s optical identity resides in first-order parameters, the study invites new forms of analysis, pedagogy, and virtual cinematography grounded in both artistic understanding and computational precision.

In this sense, the project illustrates not only the capabilities of differentiable rendering but also its potential to reshape how we understand cinematic images: not merely as artistic artifacts, but as the sum of physical choices, optical histories, and perceptual cues that can be illuminated through computation.

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